



Offline Robot Simulation Setup

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OFFLINE SIMULATION SETUP

The offline simulation setup enables the programming and simulation of robot programs without a physical connection to the robot.

The software is provided as a virtual machine (VM) image, tested exclusively with **VirtualBox version 7.1.4**. Using other versions of VirtualBox or alternative virtualization software may result in unforeseen issues. To ensure optimal performance and functionality, it is highly recommended to use **VirtualBox version 7.1.4**.

Note: This documentation is specifically designed for **VirtualBox version 7.1.4**

1.1 System Requirements

For optimal performance, your system should meet the following recommended specifications:

- **Hard Drive Space:** Minimum of 30GB of free disk space.
- **RAM:** At least 10GB is recommended for smooth operation.
- **Processor:** An 8-12 core processor is preferred for efficient performance.

Ensuring that your system meets these requirements will help avoid performance issues and provide a seamless simulation experience.

1.2 Installation Guidelines

Follow these steps to install the offline simulation software on Windows or Ubuntu. Where instructions differ for each OS, the specific steps are highlighted.

Step 1: Install VirtualBox

- **Windows:** Download and install VirtualBox from <https://www.virtualbox.org/>.
- **Ubuntu:** Download the VirtualBox .deb package from the official VirtualBox website. Once downloaded, navigate to the directory containing the file and run the following command to install it:

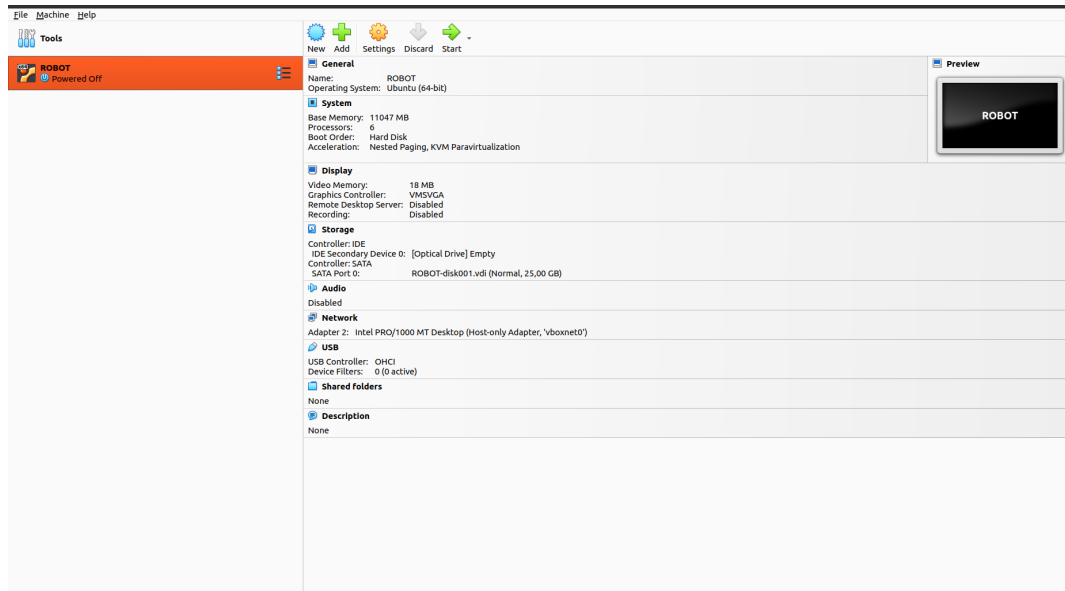
```
sudo dpkg -i ~/path/virtualbox*.deb
```

Step 2: Download the VM Image

Download the offline simulation software VM image (.ova file) to your local machine.

Step 3: Configure VirtualBox for Importing the VM

1. Open VirtualBox: - **Windows:** Navigate to the start menu and launch VirtualBox. - **Ubuntu:** Open VirtualBox from the terminal using *virtualbox*.
2. Click on the **Import** icon. In the Import Appliance window, click the folder icon beside **File**, navigate to the *.ova* file's location, and select it. Click **Open**.
3. Click **Next** to proceed. In the configuration settings, it is recommended to retain the default settings. If modified, press **Restore Defaults** before continuing.
4. Click **Import** and wait for the process to complete. An acknowledgment window will display the elapsed time. Once done, the VM will appear in the VirtualBox GUI.



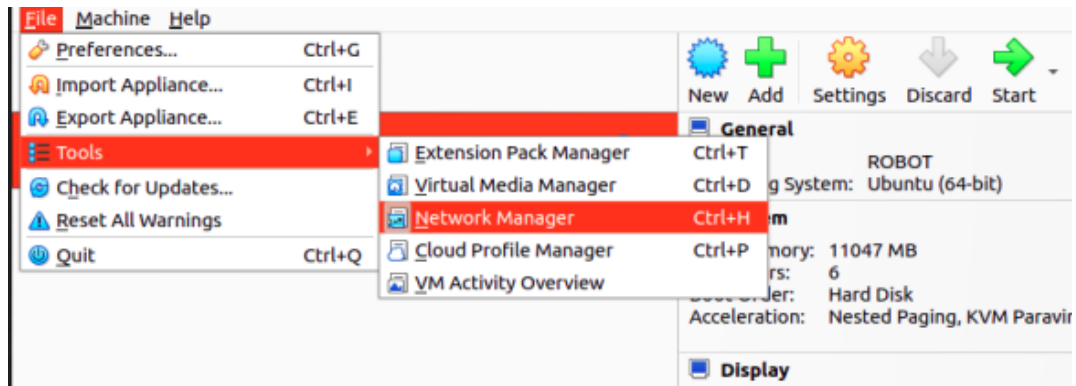
1.3 Step 4: Configure Network Settings

- **Ubuntu:** Execute following commands to enable static ip allotment to virtualbox in custom subnet

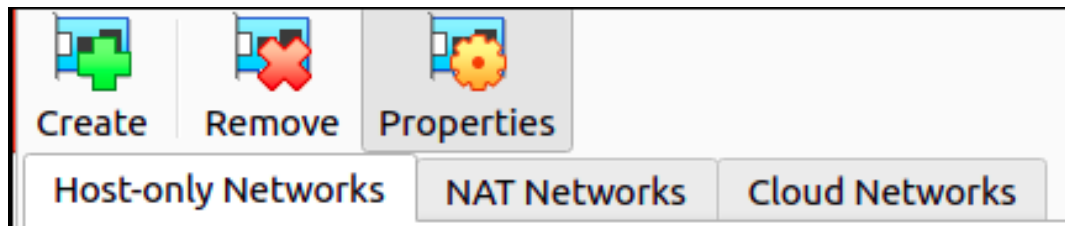
```
sudo mkdir /etc/vbox
sudo touch /etc/vbox/networks.conf
echo "* 10.0.0.0/8 192.168.0.0/16" | sudo tee -a /etc/vbox/networks.
↵conf
```

To configure the network adapter:

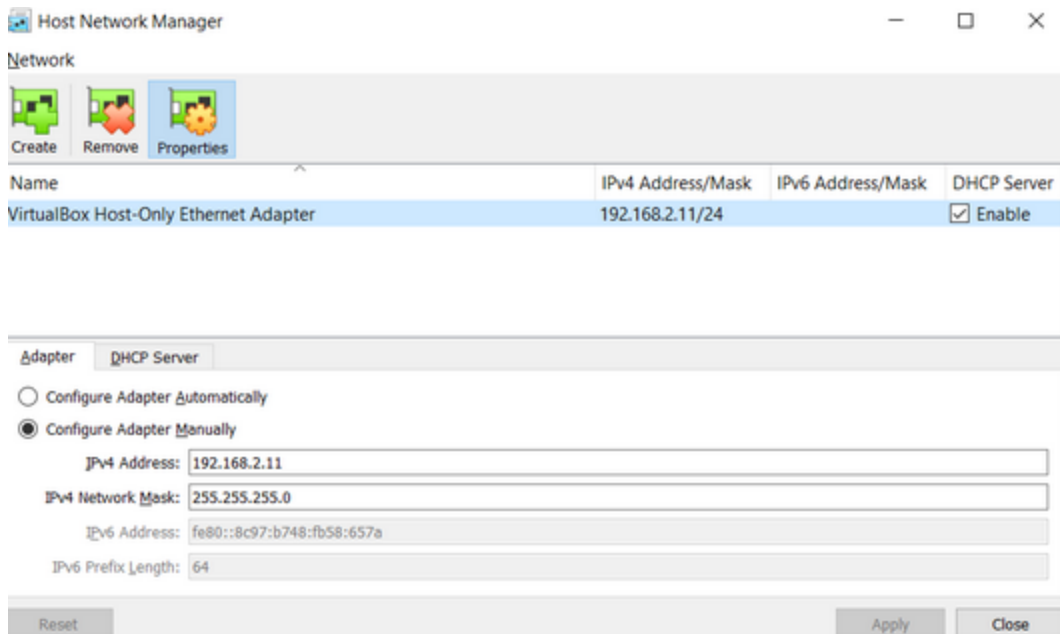
1. Navigate to **File** → **Tools** → **Network Manager** . Click **Create** to create a host-only adapter.

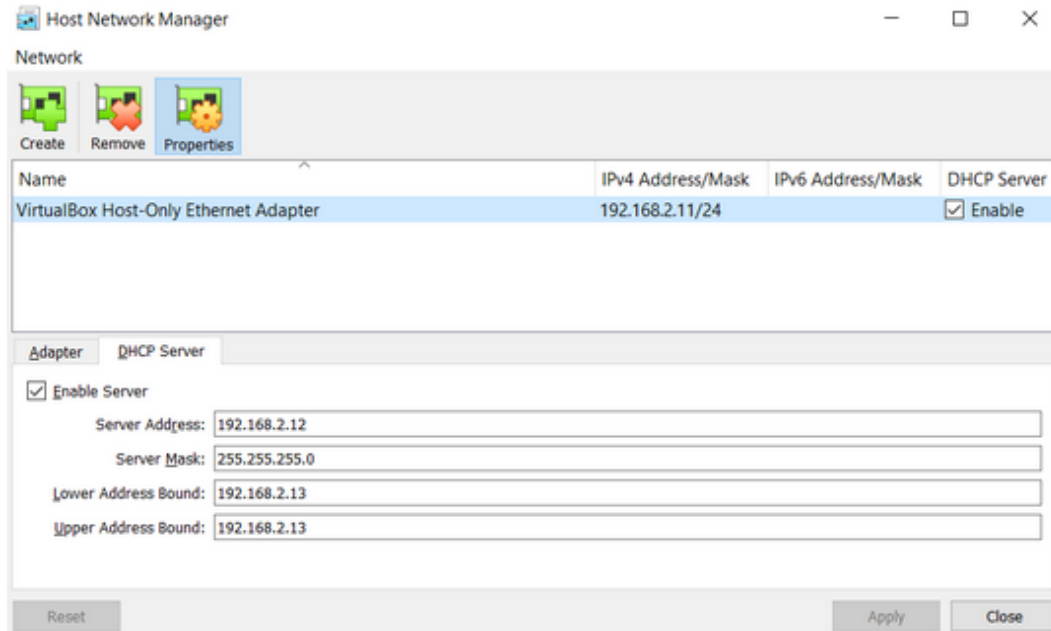


- Once the adapter is created, it should display **VirtualBox Host-Only Ethernet Adapter**.

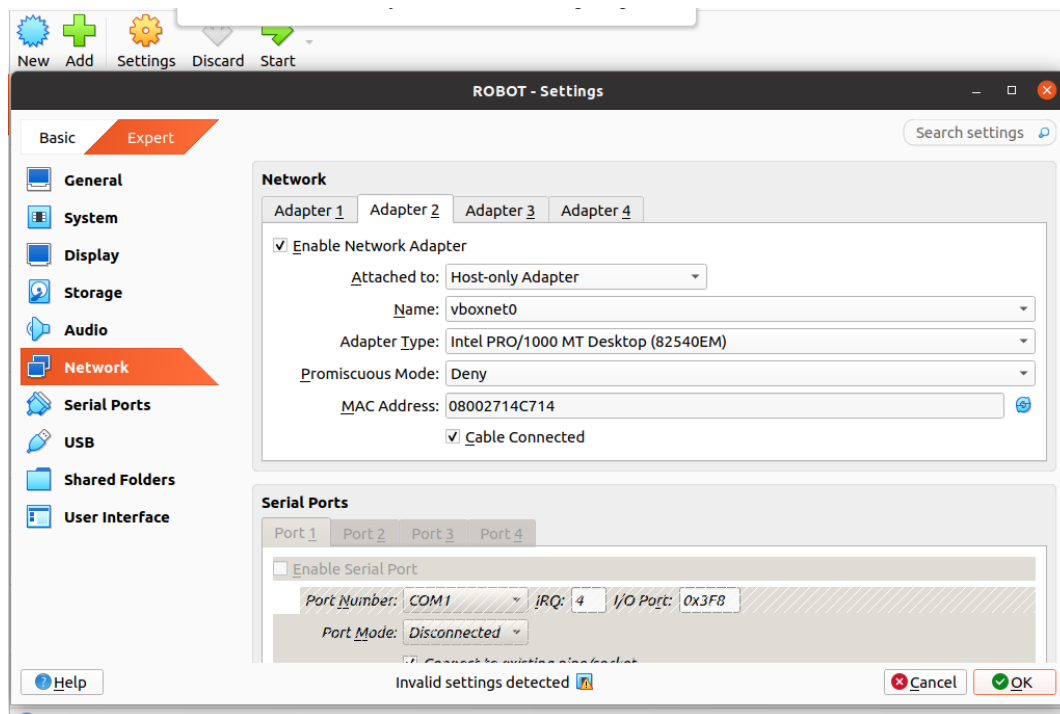


- Enable the DHCP Server by checking the **Enable** checkbox in the top right corner. Then click **Properties**.
- Choose the **Configure Adapter Manually** option and adjust the following fields: - **IPv4 Address:** 192.168.2.11 - **IPv4 Network Mask:** 255.255.255.0
- In the DHCP Server settings, configure the following: - **Server Address:** 192.168.2.12 - **Server Mask:** 255.255.255.0 - **Lower Address Bound:** 192.168.2.13 - **Upper Address Bound:** 192.168.2.13





6. Click **OK** to save the settings.
7. Navigate to the VM settings (**Settings** → **Network**) and configure the adapters as follows: - Uncheck **Enable Network Adapter** for **Adapter 1**. - Enable **Adapter 2**, set **Attached to** as **Host-only Adapter**, and select **vboxnet0**.



1.4 Step 5: Start the VM

Note: Clear the browser cache before loading the URL.

1. Start the VM and let it run for approximately 15-30 seconds.
2. Open a browser (e.g., Chrome) and enter **http://192.168.2.13:8080** or **http://192.168.2.14:8080** depending upon the Robot model. If the setup is correct, the GUI interface with the Login page should load.

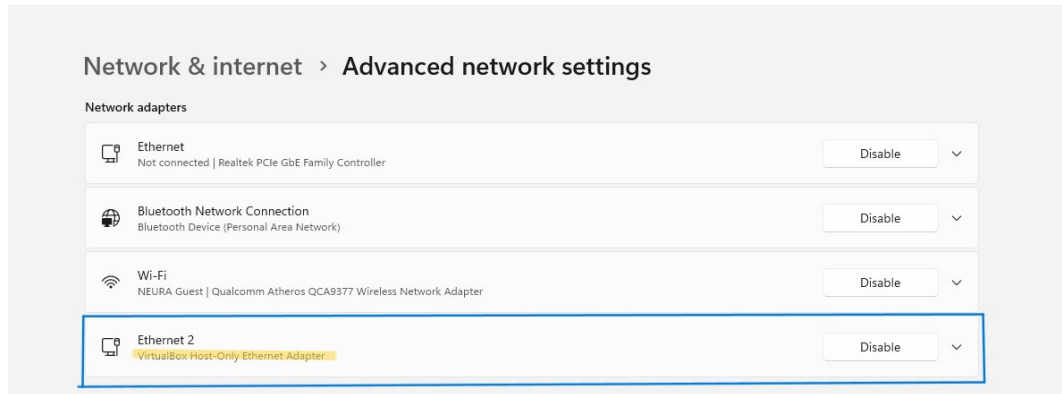
USERS

	superadmin SuperAdmin
	admin Admin
	programmer Programmer
	basicuser BasicUser

 Login

1.5 Troubleshooting Network Issues Windows

1. Verify the Ethernet Adapter's status: - **Windows:** Navigate to **Settings** → **Network & Internet** → **Advanced network settings**. - **Ubuntu:** Check the network adapter settings in the VirtualBox **Host Network Manager**.



2. Ensure the adapter is enabled. If already enabled, disable it, wait for 3 seconds, and then re-enable it.

By following these steps, you should successfully set up the offline simulation environment.

API INSTALLATION

2.1 Installation Instructions for Windows OS

2.1.1 Python 3.8 Installation

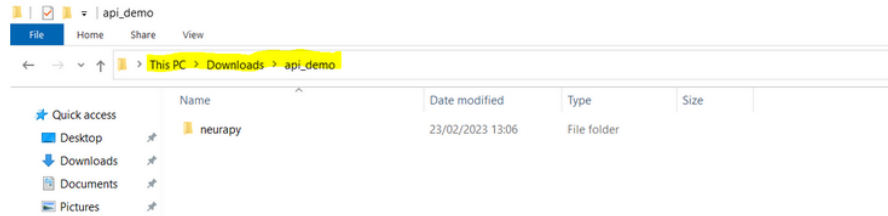
1. Download the Python 3.8 executable (.exe) file from the [official Python website](#).
2. Run the downloaded executable to begin the installation.
3. Ensure that “Add Python 3.8 to PATH” is checked before proceeding.
4. Select “Install Now” to complete the installation.



5. Close the setup once the installation is complete.

2.1.2 NeuraPy Installation

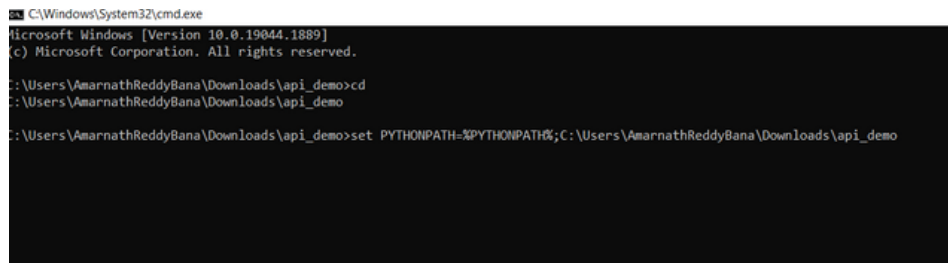
1. Download NeuraPy for Windows and navigate to the directory where it is saved.
2. Open the command prompt from this directory. You can do this by typing “cmd” in the address bar or pressing Alt+D, typing “cmd”, and then pressing Enter.



3. Install the required dependencies, *pywin32* and *prettytable*, using pip. Ensure that you have Internet connectivity for this step.

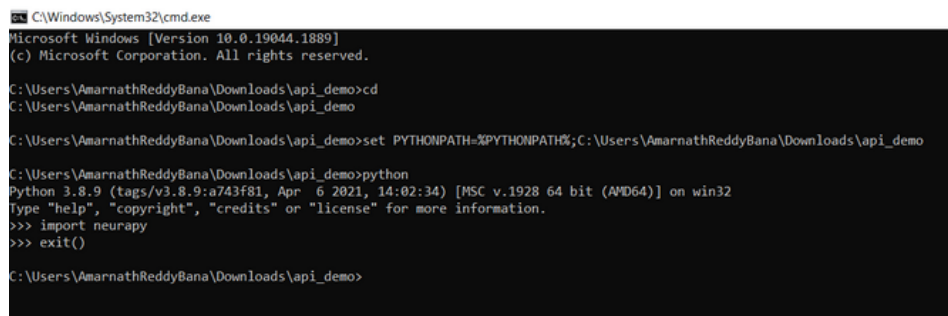
```
pip install pywin32 prettytable
```

4. Use the *cd* command in the command prompt to obtain the current location and append it to the Python path.



```
cd # Navigate to the current location
set PYTHONPATH=%PYTHONPATH%;<location obtained from previous command>
```

5. Open the Python terminal from the command prompt after setting the Python path. The API can now be imported in the Python console.



2.1.3 Configuring Client IP Address

For communication running on a non-default IP address (anything but 192.168.2.13/14), configure the IP address as follows:

1. Set an environment variable called *SOCKET_ADDRESS* to the desired IP address, e.g., 192.168.2.99.
 - Open a console by pressing the Windows key and typing *cmd*.
 - Right-click the console and select “Run as administrator.”
 - Enter the following command:

```
setx SOCKET_ADDRESS="192.168.2.99"
```

2.2 Installation Instructions for Ubuntu

Note: NeuraPy API is supported only for Ubuntu 18 and Ubuntu 20.

1. Check the Version of Ubuntu:

Open the terminal and type the following command:

```
lsb_release -a
```

2. Depending on the Ubuntu version, navigate to the *DEP_Deployment* folder of *binding_releases* and execute the *deploy_neurapy.sh* script with the respective robot name (e.g., ROBOT) as an argument:

```
cd binding_releases  
. deploy_neurapy.sh ROBOTNAME #respective robot name
```

2.2.1 Network Configuration

To control the robot via an external machine connected through Ethernet, set up the network with the following configuration:

The screenshot shows a configuration window titled "netplan-enp2s0" with tabs for Details, Identity, IPv4, IPv6, and Security. The IPv4 tab is active, showing the "IPv4 Method" set to "Manual". Under "Addresses", a table lists the IP address 192.168.2.20, netmask 255.255.255.0, and gateway 192.168.2.10. The "DNS" section has a toggle for "Automatic" set to "ON". The "Routes" section also has a toggle for "Automatic" set to "ON".

Address	Netmask	Gateway
192.168.2.20	255.255.255.0	192.168.2.10

2.2.2 Installing Visual Studio Code (VSCode)

Follow these steps to install Visual Studio Code:

1. Pre-installation Preparation:

- Operating System: Windows, macOS, or Linux.
- Internet Connection: Required for downloading the installation files.
- Storage Space: Ensure sufficient free space for the installation.

2. Downloading VSCode:

- Open a web browser (e.g., Chrome, Firefox).
- Go to <https://code.visualstudio.com/>.
- Click the download button. The website will provide the appropriate version for your operating system.

3. Installing VSCode:

- **Windows:** - Locate the downloaded *.exe* file. - Double-click to start the installation. - Follow the on-screen instructions. You can leave the settings at their default values. - Click "Install" or "Finish" to complete the installation.
- **macOS:** - Locate the downloaded *.dmg* file. - Double-click to mount the disk image. - Drag the Visual Studio Code icon to the Applications folder. - Eject the disk image after the process is complete.
- **Linux (Ubuntu):** - Open a terminal and navigate to the directory containing the *.deb* file. - Install the package using the following commands:

```
sudo dpkg -i <file-name>.deb  
sudo apt-get install -f
```

Replace *<file-name>* with the actual name of the downloaded file.

4. Getting Started:

- Launch Visual Studio Code from the applications menu or by typing *code* in the terminal.